$Y(t)$ – output

$X(t)$ – input

$$H(f) = \frac{Y(f)}{X(f)}$$

$$x(t) = \cos(2\pi t) + 10\sin(20\pi t)$$

$$H_1(j\Omega) = \frac{1}{a_1 + a_2(j\Omega)},\ H_2(j\Omega) = \frac{1}{a_3 + a_4(j\Omega) + a_5(j\Omega)^2},\ H_3(j\Omega) = e^{-ja_6\Omega}$$

$$Y_1(jw) = \frac{1}{1 + 1(jw)} \cdot X(jw)$$

$$Y_2(jw) = \frac{1}{1 + 2(jw) + 1(jw)^2} \cdot X(jw)$$

$$Y_3(jw) = e^{-10jw} \cdot X(jw)$$

$x(t)$	$X(jw)$
$\cos(2\pi t)$	$\pi[\delta(w-2\pi) + \delta(w+2\pi)]$ Re
$10\sin(20\pi t)$	$10j\pi[\delta(w+20\pi) - \delta(w-20\pi)]$ Im

$$X(jw) = \pi\left[\left(\delta(w-2\pi) + \delta(w+2\pi)\right) + 10j\left(\delta(w+20\pi) - \delta(w-20\pi)\right)\right]$$

$$Y_1(j\omega) = \frac{X(j\omega)}{s+1}$$

$$\boxed{s = j\Omega}$$

$$Y_2(j\omega) = \frac{X(j\omega)}{(s-1)^2}$$

$$Y_3(j\omega) = X(j\omega) \cdot e^{-10s}$$

$$y_1(t) = x(t) \cdot e^{-t} u(t)$$

$$x(t) = \cos(2\pi t) + 10\sin(20\pi t)$$

$$y_2(t) = x(t) \cdot t e^{t} u(t)$$

$$y_3(t) = x(t) \cdot \delta(t-10)$$